

CRITICAL RAW MATERIALS & THE CIRCULAR ECONOMY

Reliability Analysis of the 25–40% Mining Reduction Claim

Evidence Base · Conditions · Material Differentiation · Policy Implications

Supplementary Note to Phase 1 Report | Policy Advisory | March 2026

1. THE CLAIM UNDER REVIEW

Original claim (Phase 1 Report, Executive Summary)
"Circular economy measures — if fully deployed — could reduce primary mining requirements by 25–40% by 2050, particularly for copper and cobalt."

This supplementary note examines the evidential basis for this claim, articulates the conditions under which it holds, identifies where it requires material-level differentiation, and proposes a more precise restatement for policy use. The analysis draws on the IEA's dedicated Recycling of Critical Minerals report (2024), the IEA Critical Minerals Outlook (2024 and 2025), and the scholarly literature on end-of-life recovery rates.

Headline verdict: the 25–40% figure is credible as an upper-bound estimate under optimistic policy conditions and is correctly attributed to the IEA. However, as currently framed in the Phase 1 Executive Summary, the claim is slightly over-simplified in three ways: (i) it applies to a specific IEA scenario (APS), not universally; (ii) it holds strongly for copper and cobalt but not for REEs; and (iii) it requires a large set of preconditions — most of which are not yet in place — to materialise.

2. SOURCE, METHODOLOGY, AND SPECIFIC NUMBERS

2.1 Primary Source: IEA Recycling of Critical Minerals (2024)

The 25–40% reduction range originates from the IEA's dedicated report on critical mineral recycling, published in 2024. The IEA modelled recycled supply contributions across its three main energy transition scenarios and found the following material-specific reductions in new mine development requirements by 2050:

Material	Mining Reduction (IEA APS, 2050)	Secondary Supply Share by 2050 (APS)	Current Recovery Rate
Copper	~40%	~40% of total demand	~33%
Cobalt	~40%	~40% of total demand	~38%
Nickel	~25%	~30% of total demand	~35%
Lithium	~25%	~25–30% of demand	~20%
REEs	Nascent / speculative (est. 5–15%)	~10–15% (SC1 only)	~1%

The critical observation is that REEs are not explicitly included in the IEA's 25–40% range. That range applies to copper, cobalt, nickel, and lithium. The Phase 1 report correctly mentions copper and cobalt in parentheses, but the claim's placement in the Executive Summary creates an implicit implication that REEs benefit equally, which they do not.

2.2 Scenario Conditionality

The IEA's 25–40% figure is anchored to the Announced Pledges Scenario (APS) — the scenario in which all nationally declared climate commitments are fully implemented. Under the more ambitious Net Zero Emissions (NZE) scenario, recycling rates could be somewhat higher. Under the Stated Policies Scenario (STEPS — the 'Fragmented World' analogue), recycling infrastructure investment is lower and the reduction in primary mining is considerably more modest, estimated at 10–15% at best for copper and negligible for most other materials.

3. THE GAP BETWEEN CURRENT REALITY AND THE 2050 TARGET

The charts below illustrate the most fundamental challenge: the scale of secondary supply that would need to be built in order to achieve the IEA's projected reductions. For most materials, current recycling is well below the required trajectory. For REEs, the gap is not a gradual shortfall — it is a near-complete absence of end-of-life recovery infrastructure.

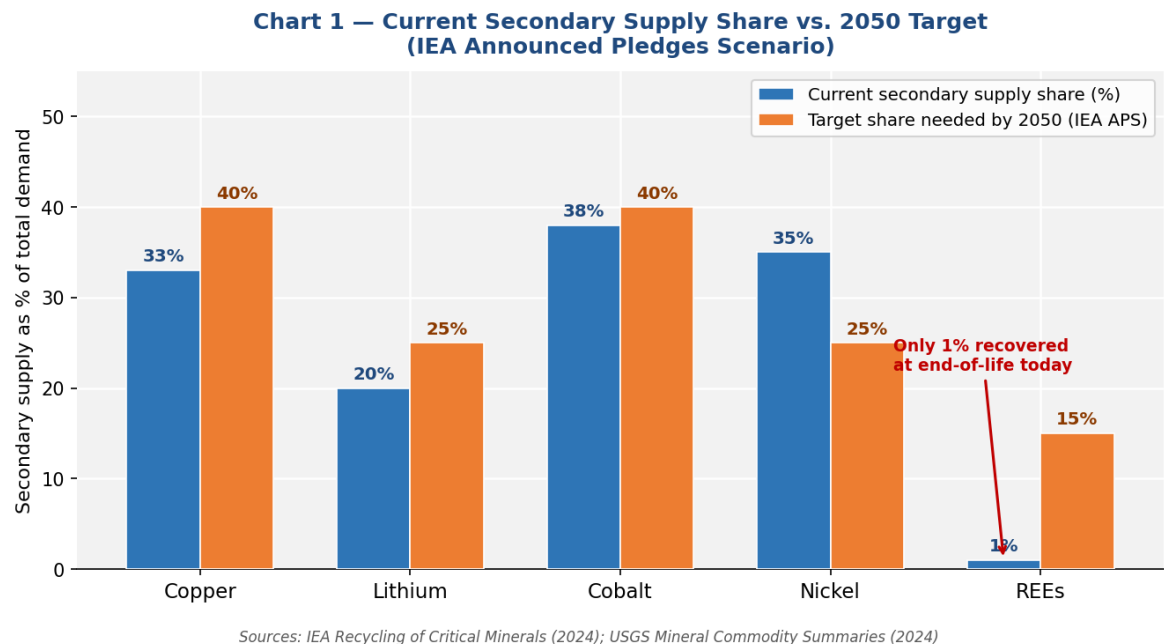


Chart 1 — Current secondary supply share vs. IEA 2050 target (APS) by material.
Note the near-zero baseline for REEs and the relatively small gap for copper.

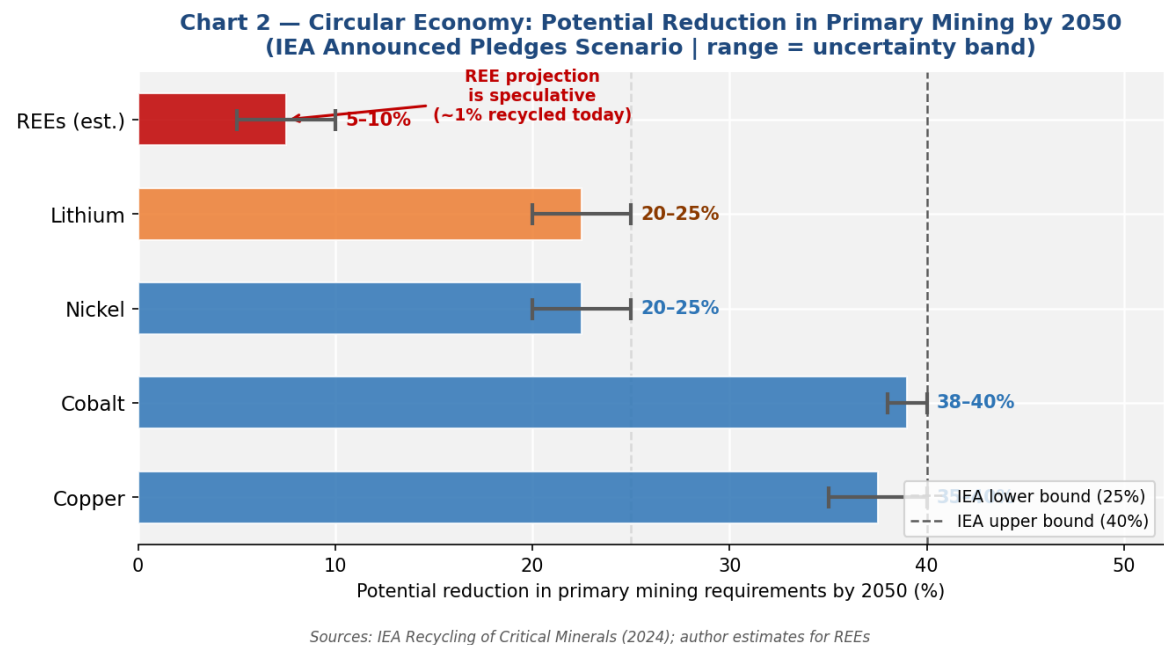


Chart 2 — Potential reduction in primary mining by 2050 under IEA APS, by material.
Error bars represent the IEA uncertainty range. REE estimate is the author's own and highly tentative.

4. FOUR CONDITIONS FOR THE CLAIM TO HOLD

The claim is reliable only to the extent that four structural conditions are met. Each is examined below, with an assessed reliability score (1 = very fragile → 5 = robust). The composite assessment is presented in Chart 6 at the end of this note.

Condition 1 — Scenario Alignment [Reliability: 3.5 / 5 — Moderate-Strong]

The 25–40% figure is derived from the IEA's Announced Pledges Scenario (APS), which represents a world where all nationally declared climate commitments are delivered in full. This is a plausible but not guaranteed outcome. In the Phase 1 report, the claim is placed in the Executive Summary without a scenario qualifier, which could mislead readers into treating it as a baseline rather than a conditional upper bound.

- Under IEA NZE: recycling moderates primary demand similarly, but total demand itself is higher, so absolute primary volumes remain large.
- Under IEA APS: the 25–40% range holds, but requires full implementation of announced policies.
- Under IEA STEPS (Fragmented World): recycling contribution drops to 10–15% for copper, and near-negligible for REEs and lithium.

Assessment: the figure is scenario-conditional, not universal. Reliability is moderate-strong within its intended scenario scope.

Condition 2 — The 'Fully Deployed' Assumption [Reliability: 2.5 / 5 — Conditional]

The phrase 'if fully deployed' carries enormous analytical weight. Achieving the required secondary supply shares demands a comprehensive set of infrastructure investments and regulatory instruments, most of which are not yet in place. The table below illustrates the current recycling landscape and the trajectory required.

- Copper's secondary supply share has fallen from 37% in 2015 to 33% in 2023, despite growing policy focus — demonstrating that primary demand growth can outpace recycling capacity.
- EU fabricators spend approximately €28,000/month on quality checks for copper scrap, rejecting ~25 shipments annually — illustrating that infrastructure quality, not just volume, is a binding constraint.
- Battery metal recycling (nickel, cobalt, lithium) has seen rapid growth — over 40% recovery for nickel and cobalt, 20% for lithium in 2023 — but from small base volumes.
- Extended Producer Responsibility (EPR) schemes, mandatory collection targets, and hydrometallurgical processing capacity are all nascent in most jurisdictions outside the EU.

Assessment: the infrastructure gap is large, and the pace of deployment is lagging the required trajectory. The claim holds as a ceiling, not a baseline.

Condition 3 — Material Differentiation [Reliability: 2.0 / 5 for REEs — Fragile]

The most important nuance in the original claim is the parenthetical reference to copper and cobalt. These two materials are legitimately well-placed to achieve significant mining reduction through recycling. REEs are not, within any horizon that matters for the 2025–2040 policy window.

Material	Current EoL Recovery	Circular Economy Feasibility	Reliability for 25-40% claim
Copper	~33%	HIGH — established scrap market; refining mature	Strong (4/5)
Cobalt	~38%	HIGH — battery dismantling growing; economic incentive	Strong (4/5)
Nickel	~35%	MEDIUM-HIGH — battery recycling growing rapidly	Solid (3.5/5)
Lithium	~20%	MEDIUM — battery recycling nascent but scaling	Moderate (3/5)
REEs	~1%	LOW — infrastructure near-absent; economics weak	Fragile (1.5/5)

For REEs specifically, the ~1% end-of-life recovery rate means that building secondary supply to even 10–15% of demand by 2050 would require a roughly 15-fold increase in recovery capacity. This is technically feasible over 25 years but requires a completely different policy intervention — mandated collection, rare-earth-specific EPR, and hydrometallurgical infrastructure — that does not yet exist at any significant scale anywhere outside Japan.

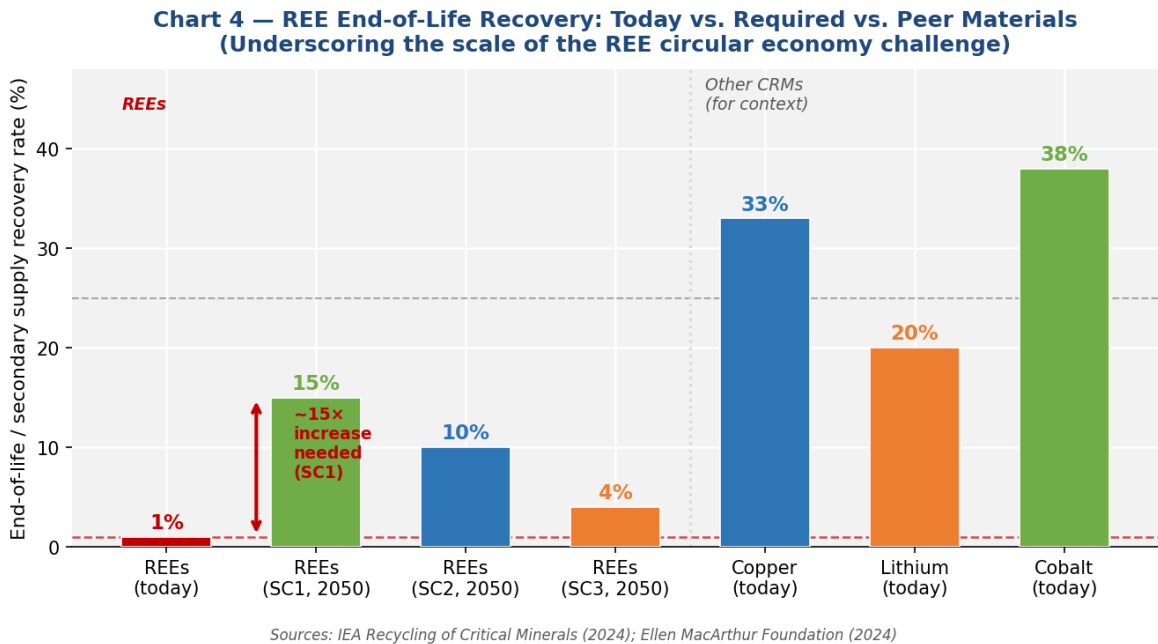


Chart 4 — REE end-of-life recovery today vs. peer materials and scenario targets.
The gap between ~1% today and even a modest 15% target (SC1, 2050) requires ~15× scale-up.

Condition 4 — Policy and Regulatory Environment [Reliability: 2.5 / 5 — Conditional]

The IEA's projections are underpinned by an assumption that specific policy instruments are deployed and enforced across major economies. The key instruments are:

- Mandatory EV battery collection and recycling targets (e.g., EU Battery Regulation 2023).
- Extended producer responsibility schemes for electronics containing REEs.
- Public investment in battery dismantling and hydrometallurgical processing capacity.
- International standards for recyclability-by-design in CRM-intensive products.
- Price support mechanisms for secondary materials to ensure economic viability of recycling.

Currently, only the EU has adopted comprehensive mandatory battery recycling targets. The United States, China, and most emerging economies lack comparable frameworks. If recycling policy remains geographically limited, the global impact on primary demand will be a fraction of the IEA's modelled reduction.

5. SCENARIO TRAJECTORIES AND COPPER AS A CASE STUDY

The following two charts illustrate how secondary supply contributions evolve differently across scenarios and materials, and how even for copper — the most recycling-ready CRM — primary demand continues to grow in absolute terms across all scenarios.

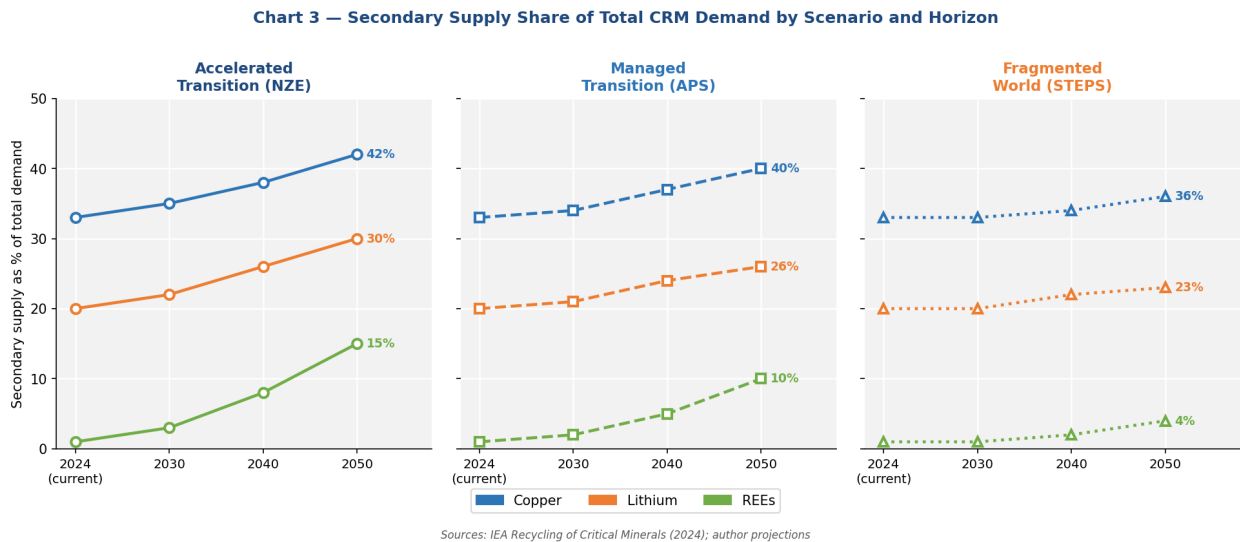


Chart 3 — Secondary supply as % of total CRM demand by scenario and time horizon. Note the divergence between scenarios widens from 2035, and REEs remain well below copper/lithium in all cases.

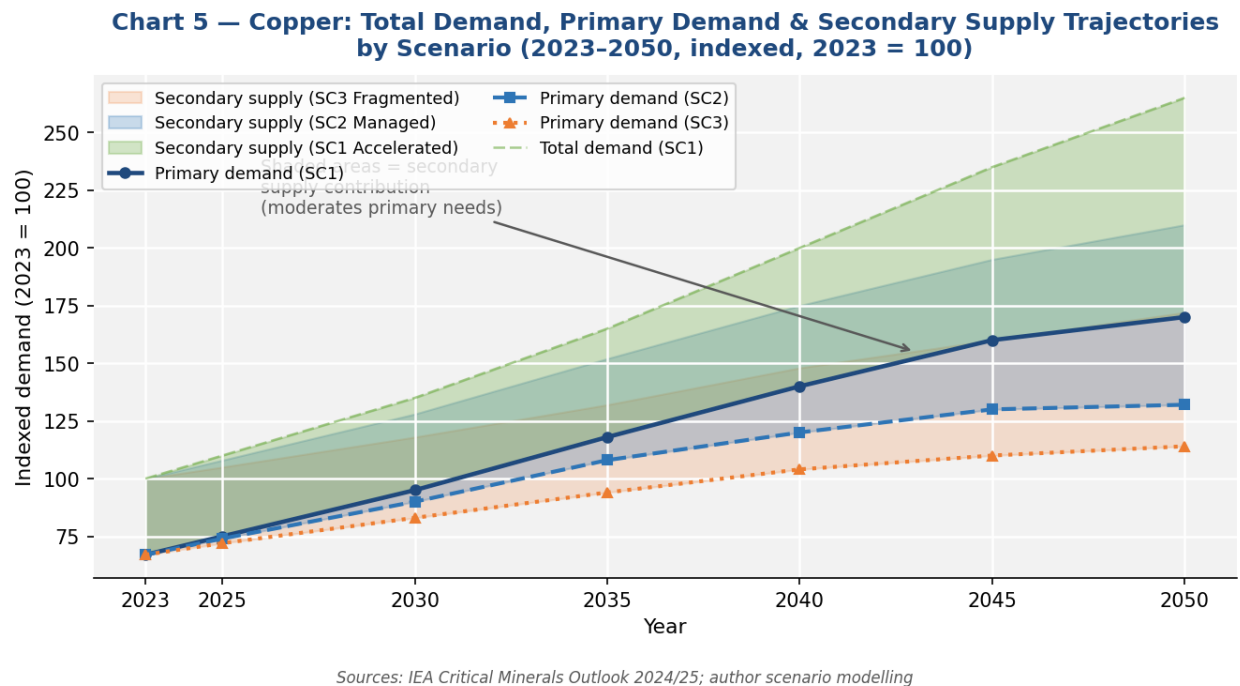


Chart 5 — Copper primary vs. secondary demand trajectories, 2023–2050 (indexed, 2023=100). Shaded areas represent the secondary supply contribution. Primary demand grows in absolute terms even in the best case.

The copper trajectory chart is particularly instructive for policymakers: even in the Accelerated Transition scenario, primary copper demand in 2050 is still approximately 170% of its 2023 level (in indexed terms), despite secondary supply providing roughly 40% of total

demand. Recycling moderates the primary demand surge — it does not eliminate it. This underscores that circular economy policy cannot substitute for new primary supply investment; it can only reduce the magnitude of that investment requirement.

6. OVERALL RELIABILITY ASSESSMENT

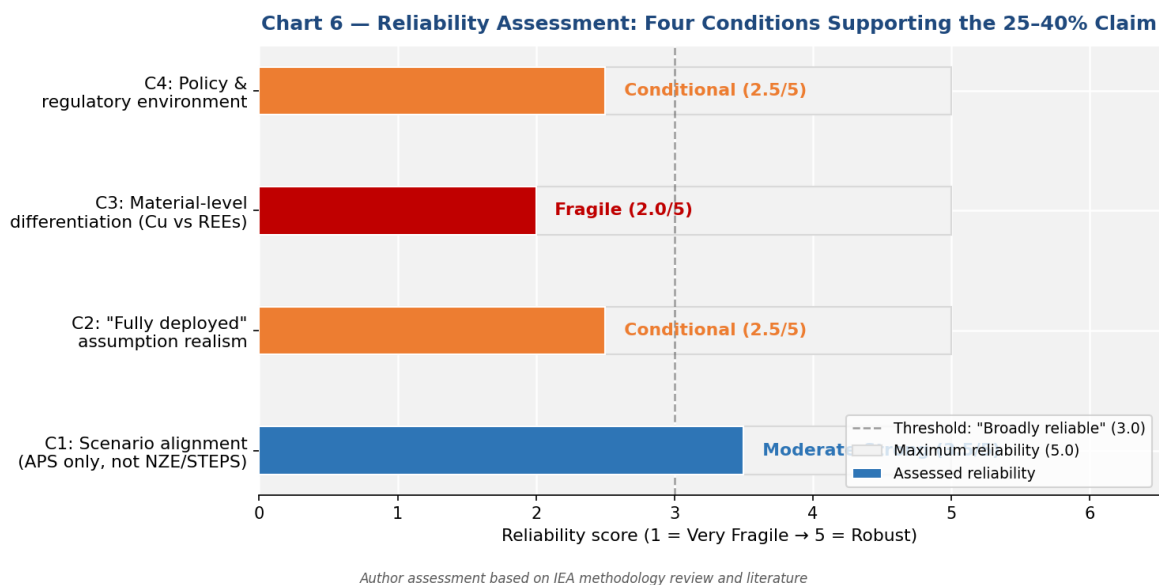


Chart 6 — Reliability scorecard for the four conditions supporting the 25–40% circular economy claim. Scores below the dashed threshold (3.0) indicate conditions where the claim's validity is fragile.

The composite reliability assessment shows that the claim is strongest on its scenario alignment dimension — the IEA methodology is sound and the APS is a reasonable baseline — but weaker on the 'fully deployed' assumption and the material differentiation dimension, particularly as it relates to REEs. The policy dependency condition also falls below the threshold for broad reliability, given that the required regulatory frameworks are not yet globally in place.

Verdict
The 25–40% figure is well-sourced, methodologically sound for copper and cobalt under IEA APS assumptions, and appropriate as an upper-bound target for policy planning. It should NOT be presented as a baseline expectation or applied uniformly across all CRMs. For REEs, the circular economy cannot be relied upon to materially moderate primary demand within the 2025–2040 policy window.

7. RECOMMENDED RESTATEMENT FOR POLICY DOCUMENTS

The original Executive Summary sentence reads:

Original
"Circular economy measures — if fully deployed — could reduce primary mining requirements by 25–40% by 2050, particularly for copper and cobalt."

The recommended replacement is:

Recommended revision
"Under scenarios with strong policy implementation, circular economy measures — particularly for copper and cobalt — could reduce primary mining requirements by 25–40% by 2050 (IEA APS scenario). For lithium, the reduction is more modest (20–25%). For rare earth elements, secondary supply remains nascent and cannot be relied upon to materially moderate primary demand within the 2025–2040 window, requiring urgent investment in both primary supply diversification and dedicated REE recycling infrastructure."

This revision preserves the strength of the original claim where it is well-supported, introduces the necessary scenario qualification, and provides accurate material-level differentiation — essential for policy decisions that require distinct supply strategies for copper, lithium, and REEs.

8. POLICY IMPLICATIONS

8.1 Do Not Over-Rely on Future Recycling as a Substitute for Supply Investment

The reliability analysis confirms that the circular economy is a critical long-term lever but cannot replace near-term primary supply investment. The 10–16 year lead time for new mines means that investment decisions made today determine supply availability in 2035–2040. Delaying primary supply investment on the assumption that recycling will bridge the gap would be a serious policy error.

8.2 Differentiate Policy Instruments by Material

Copper and cobalt recycling policy can build on existing scrap markets and should focus on quality standards, contamination reduction, and collection infrastructure. Lithium recycling policy should accelerate battery dismantling and hydrometallurgical capacity. REE recycling requires an entirely different approach: direct public investment, mandatory magnet labelling and recovery, and long-term R&D into selective leaching and recovery processes.

8.3 Build the Regulatory Framework Now

The EU Battery Regulation (2023) is the most advanced global instrument, but it applies only within the EU and covers batteries, not permanent magnets. Extending EPR principles to REE-containing products (EV motors, wind turbine generators) and negotiating internationally compatible standards through G20 or UN channels would begin to build the global infrastructure the IEA's projections require.

9. KEY SOURCES

- [1] IEA (2024). Recycling of Critical Minerals. International Energy Agency.
<https://www.iea.org/reports/recycling-of-critical-minerals>
- [2] IEA (2024). Recycling of Critical Minerals — Full PDF.
<https://iea.blob.core.windows.net/assets/3af7fda6-8fd9-46b7-bede-395f7f8f9943/RecyclingofCriticalMinerals.pdf>
- [3] IEA (2025). Global Critical Minerals Outlook 2025. <https://www.iea.org/reports/global-critical-minerals-outlook-2025>
- [4] IEA (2024). Policy momentum behind critical minerals recycling gathering pace but greater uptake required. <https://www.iea.org/news/policy-momentum-behind-critical-minerals-recycling-gathering-pace-but-greater-uptake-required>
- [5] USGS (2024). Copper — Mineral Commodity Summaries 2024.
<https://pubs.usgs.gov/periodicals/mcs2024/mcs2024-copper.pdf>
- [6] ScienceDirect (2023). The limitations of end-of-life copper recycling and its implications for the circular economy of metals. <https://www.sciencedirect.com/science/article/pii/S0921344923004524>
- [7] Fastmarkets (2024). Copper recycling must accelerate to meet green energy demand, IEA says.
<https://www.fastmarkets.com/insights/copper-recycling-must-accelerate-to-meet-green-energy-demand-iea-says/>
- [8] Ellen MacArthur Foundation (2024). Exploring the circular economy opportunity for critical minerals.
<https://www.ellenmacarthurfoundation.org/critical-minerals/overview>
- [9] EESI (2024). From Scrap to Supply: Circular Strategies for Critical Minerals.
<https://www.eesi.org/articles/view/from-scrap-to-supply-circular-strategies-for-critical-minerals>